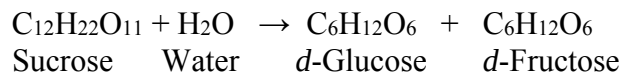


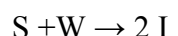
(10 pts.) Be sure to print your ABET Survey Code at the top of your solutions!

SPECIAL NOTE: Do all calculations on your own paper. To receive full credit, write your final answer either in the space provide on this sheet or clearly on your own sheet. Send an image of your solution to me at wedge@uic.edu in a single PDF file. Name your PDF file "Your last name" "Your first name" Final Exam.

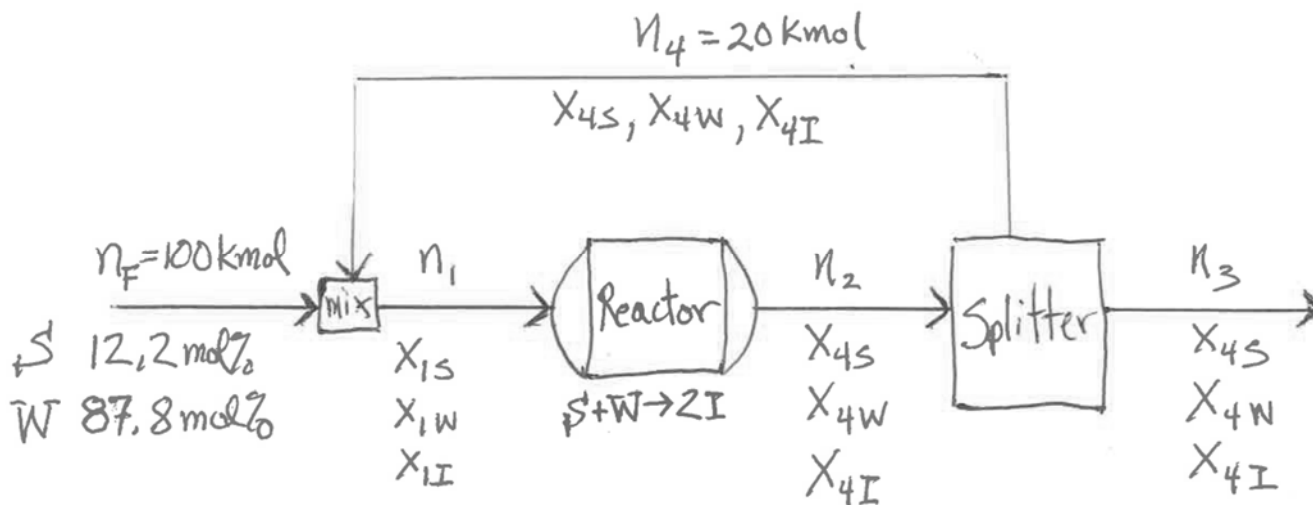
1. (50 points) Sucrose ($C_{12}H_{22}O_{11}$) can be converted to glucose and fructose (both $C_6H_{12}O_6$) according to the reaction



The mixture of glucose and fructose is called inversion sugar (I). If we give sucrose the symbol S and water the symbol W, the reaction can be written simply as



The process is shown below where the reactor has a single-pass conversion of sucrose of 80%. The process is fed 100 kmol of 12.2 mol% S and the remainder W. The reactor effluent is fed into a splitter with a product stream and a recycle stream of identical composition. The recycle stream has a flow rate of $n_4 = 20$ kmol.



- (a) Do a degree of freedom analysis following page 105 of the text and show that the problem is sufficiently specified. (5 pts.)

- (b) Using balances on the total molar flow rates, calculate the molar flow rates of the reactor input, output, and the product stream. (15 pts.)

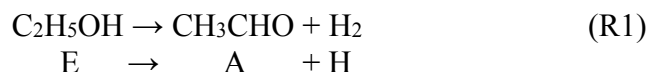
$n_1 =$ _____ kmol input, $n_2 =$ _____ kmol output, $n_3 =$ _____ kmol product

- (c) Using balances on species, write six more independent equations to calculate the mole fractions of the input and output stream for the reactor. These equations should be linear in the unknowns. (30 pts.)

$X_{1S} =$ _____, $X_{1W} =$ _____, $X_{1I} =$ _____

$X_{2S} =$ _____, $X_{2W} =$ _____, $X_{2I} =$ _____

2. (65 points) Acetaldehyde (CH_3CHO) is produced by dehydrogenation of ethanol ($\text{C}_2\text{H}_5\text{OH}$).



An undesired side reaction produces ethyl acetate ($\text{CH}_3\text{COOC}_2\text{H}_5$) according to the reaction



The steady molar flow rate into the reactor is 120 mol/s of pure ethanol gas at 200°C and 1 atm. The total molar flow rate of the stream exiting the reactor is 166 mol/s containing ethanol (E), hydrogen (H), acetaldehyde (A), and ethyl acetate (Ea) also at 200°C and 1 atm. All components of the exit stream are gases. To maintain this isothermal condition, 2228 kJ/s of heat is added to the reactor.

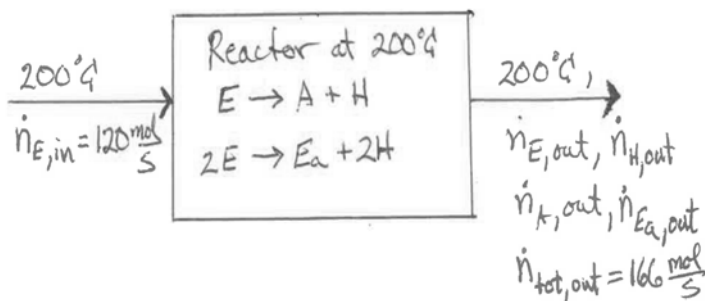
Data:

Ethanol: $C_p(\text{J/mol K}) = 21.96 + 0.1477 T(\text{K})$

Acetaldehyde: $C_p(\text{J/mol K}) = 61.00$

Hydrogen: $C_p(\text{J/mol K}) = 29.20$

Ethyl Acetate: $C_p(\text{J/mol K}) = 42.59 + 0.2397 T(\text{K})$



- (a) Write species balances on the reactor in terms of the extents of reaction for both reactions, that is, ξ_1 and ξ_2 . Also give the change in total moles over the reactor in terms of ξ_1 and ξ_2 . (15 pts.)
- (b) Calculate the enthalpy of reaction for both reaction as written using enthalpies of formation from the textbook. (10 pts.)

$$\Delta \hat{H}_{R1}^\circ = \underline{\hspace{2cm}} \text{ kJ/mol}, \quad \Delta \hat{H}_{R2}^\circ = \underline{\hspace{2cm}} \text{ kJ/mol}$$

- (c) Write an energy balance on the reactor ignoring shaft work and mechanical energy. Combine the resulting energy balance with the result from Part (a) for the total mole balance to determine ξ_1 and ξ_2 for both reactions. Use the heat capacity data given above rather than in the text. (20 pts)

$$\xi_1 = \underline{\hspace{2cm}} \text{ mol/s}, \quad \xi_2 = \underline{\hspace{2cm}} \text{ mol/s}$$

- (d) Calculate the species molar flow rates of the exiting stream from the reactor in mol/s. Complete the table below including the enthalpies at 200 °C. (10 pts.)

Species	$\dot{n}_{exit}$ (mol/s)	$\hat{H}(200^\circ\text{C})$
Ethanol (E)		
Acetaldehyde (A)		
Hydrogen (H)		
Ethyl Acetate (Ea)		

- (e) Calculate the fractional conversion of ethanol and the fractional yield of acetaldehyde. (10 pts.)

$$f_{C, \text{Ethanol}} = \underline{\hspace{2cm}}, \quad \text{fractional yield} = \underline{\hspace{2cm}}$$

3. (35 points) Recall Problem 1 from Exam #2:

“Consider a binary liquid mixture of species A and B. A storage tank initially contains 100 kg of a 60 wt% of A in the mixture. That is, the weight fraction of the A in the tank is initially $w = 0.6$. Starting at time $t = 0$, a 10 wt% of A, $w_i = 0.1$, is added at a constant rate of $\dot{m}_i = 10 \text{ kg/min}$ while at the same time the exit flow rate is proportional to the current mass in the tank such that $\dot{m}_e = k m_{sys}$. The constant $k = 0.15 \text{ min}^{-1}$ so that the initial exit flow rate is $\dot{m}_e(0) = 15 \text{ kg/min}$. Assume that the tank is well mixed so that the A weight fraction at the exit is the same as the weight fraction in the tank, $w_e = w(t)$.”

In addition, there is a reaction taking place in the tank such that $A \rightarrow 2B$ where the reaction rate for the disappearance of A in kmol/min is given by $r_A = K V c_A$ where $K = 5.2 \text{ min}^{-1}$ is the reaction rate constant, V is the volume of liquid in the tank in m^3 , and c_A is the molar concentration of A in the tank kmol/m^3 . The molecular weights of A and B are 52 kg/kmol and 26 kg/kmol respectively, and the density of both A and B is 34 kg/m^3 . From Exam #2 we already know that the mass in the tank as a function of time is given by:

$$m_{sys} = \frac{10 + 5e^{-0.15t}}{0.15}$$
 where t is in minutes and m_{sys} is in kg and $m_{sys} = 74.10 \text{ kg}$ after 10 minutes. Because the densities of both A and B are equal, this result is still valid.

What is the mass fraction w after the same 10 minutes?

